Theme-park management system

Modules Logic + Test Cases

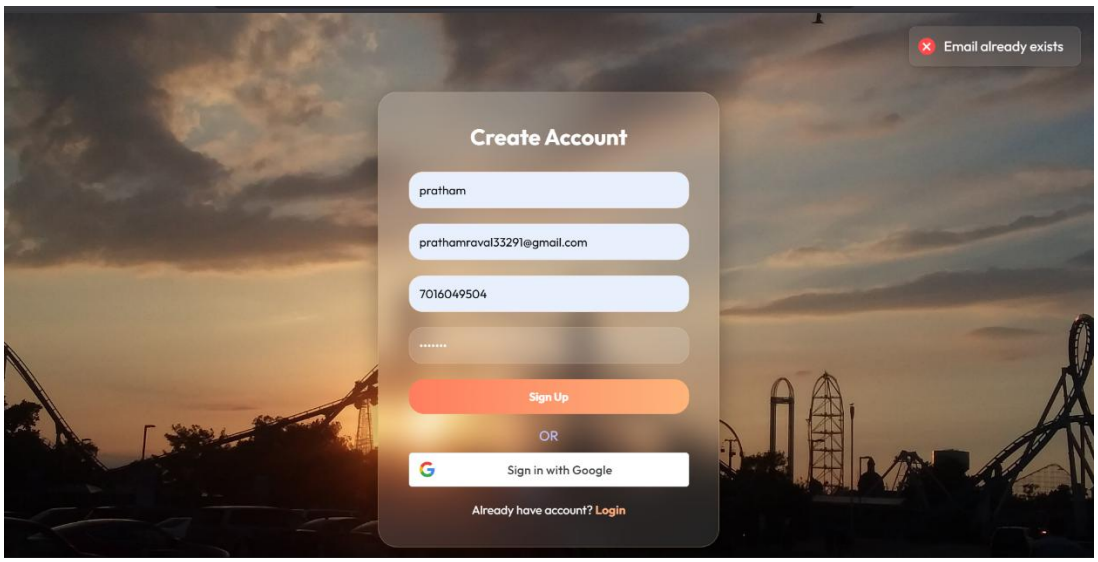
1. Student Profile & Skill Management Module

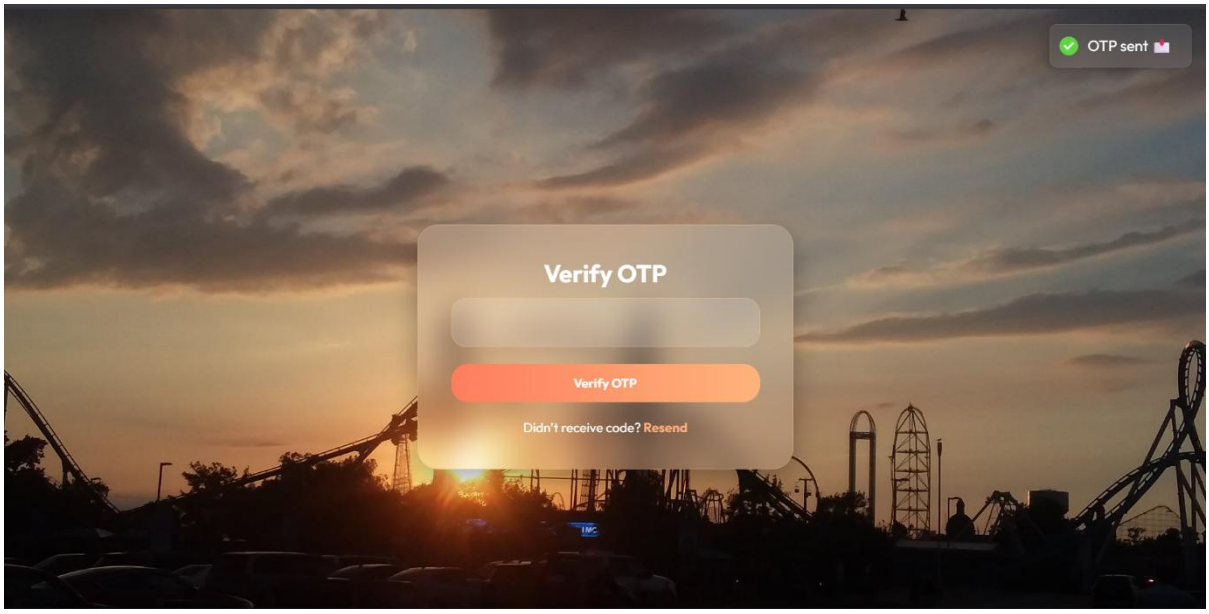
1.1. User enters signup details
1.2. System validates input
1.3. Password is hashed using bcrypt
1.4. Data stored in MongoDB
1.5. User logs in using credentials
1.6. Role-based access applied

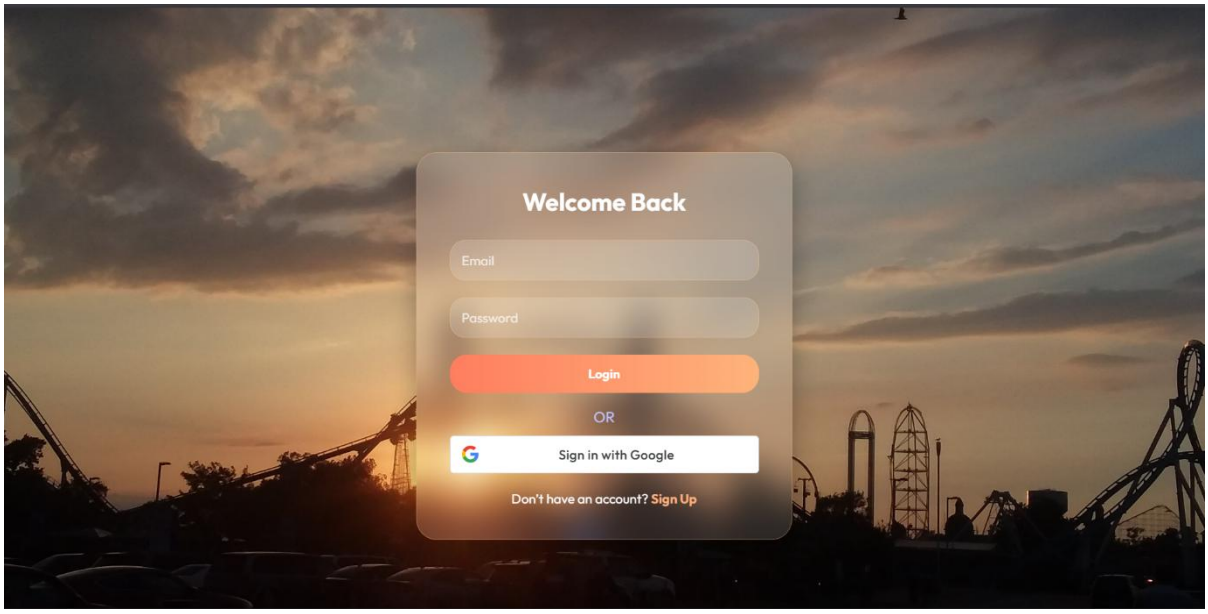
Test Cases:

TC ID	Scenario	Input	Expected Output
TC01	User signup	Valid details	Account created
TC02	Duplicate email	Existing email	Error message
TC03	Login	Correct credentials	Login success
TC04	Invalid login	Wrong password	Error shown
TC05	Role Validation	User Role	Role Based Page Redirecting

ScreenShots :




OTP sent
Verify OTP
Verify OTP
Didn't receive code? Resend


Welcome Back
Email
Password
Login
OR
Sign in with Google
Don't have an account? Sign Up

2. Google Authentication

2.1 User clicks “Login with Google”

2.2 Google OAuth token is generated

2.3 Backend verifies token using google-auth-library

2.4 If user exists → login

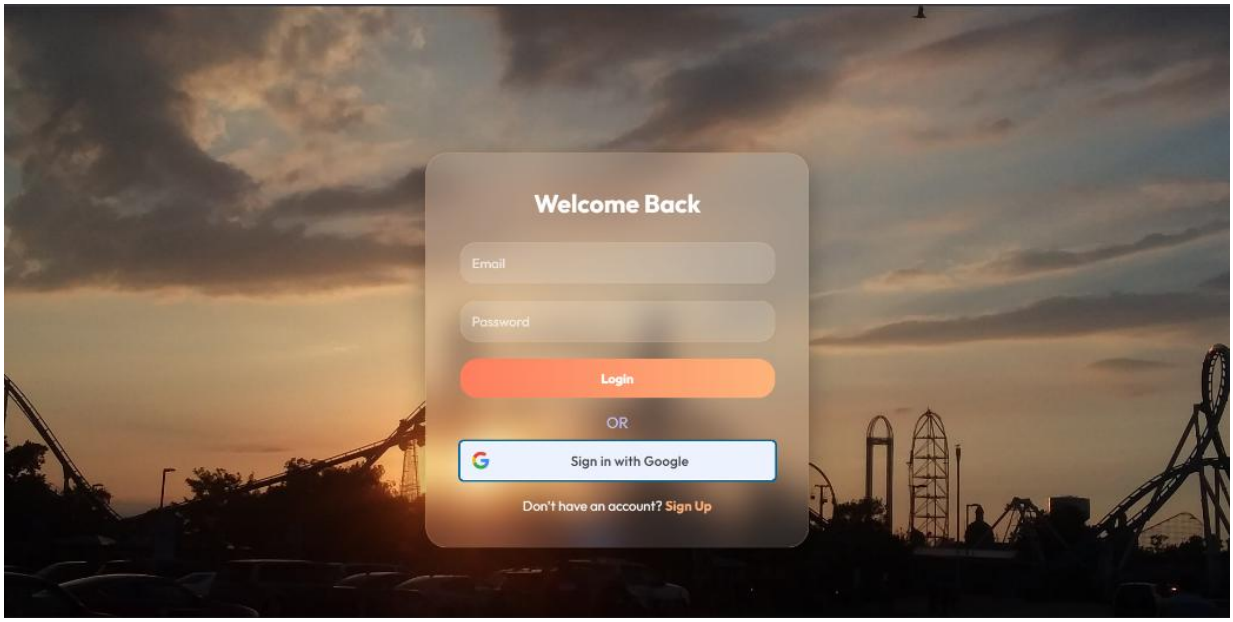
2.5 Else → new user created

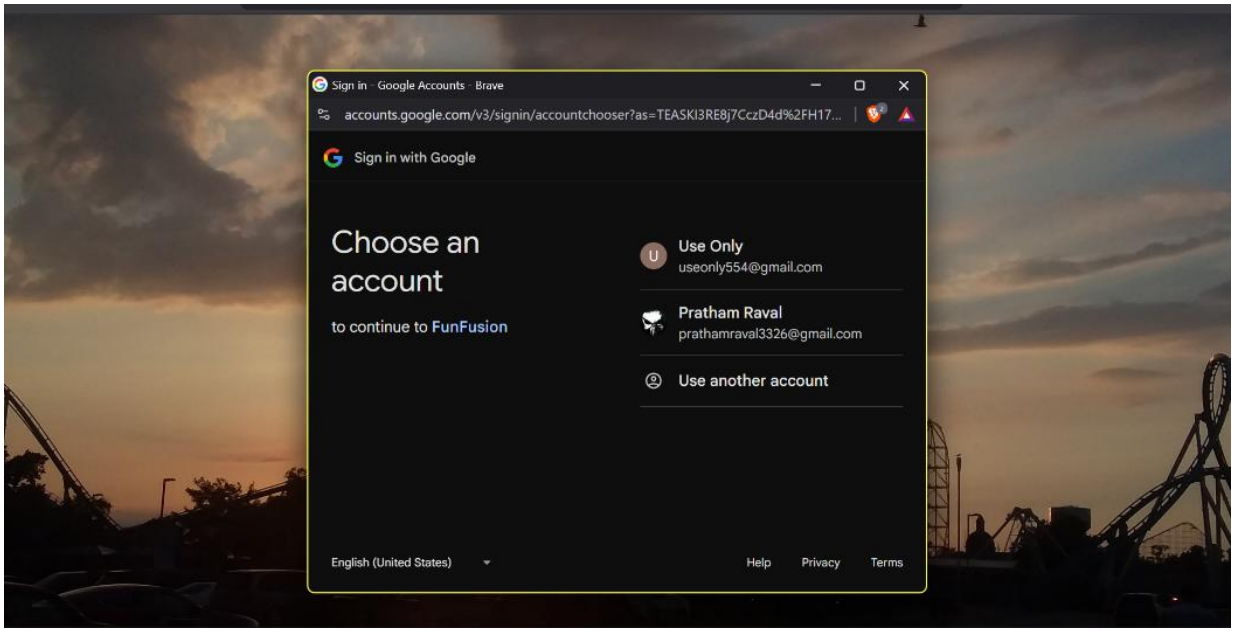
2.6 User is redirected to dashboard

Test Cases:

TC ID	Scenario	Input	Expected Output
TC06	Google login	Valid token	Login successful
TC07	New user	First-time login	Account created
TC08	Existing user	Already registered	Direct login

ScreenShots :





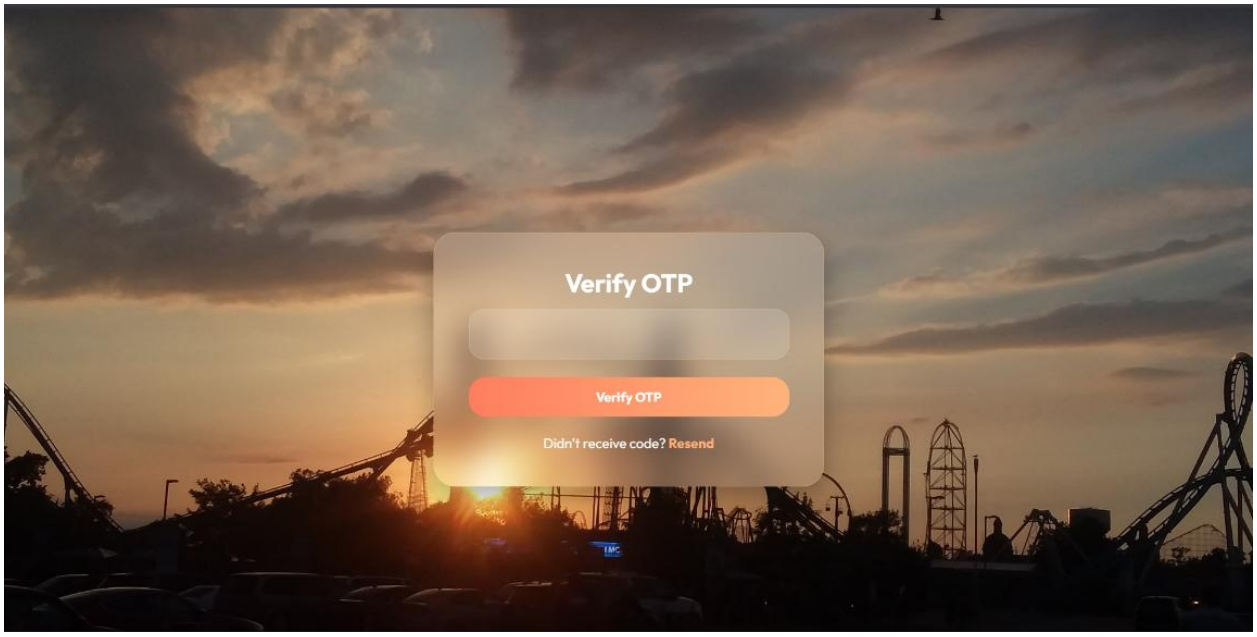
3. OTP Verification Module

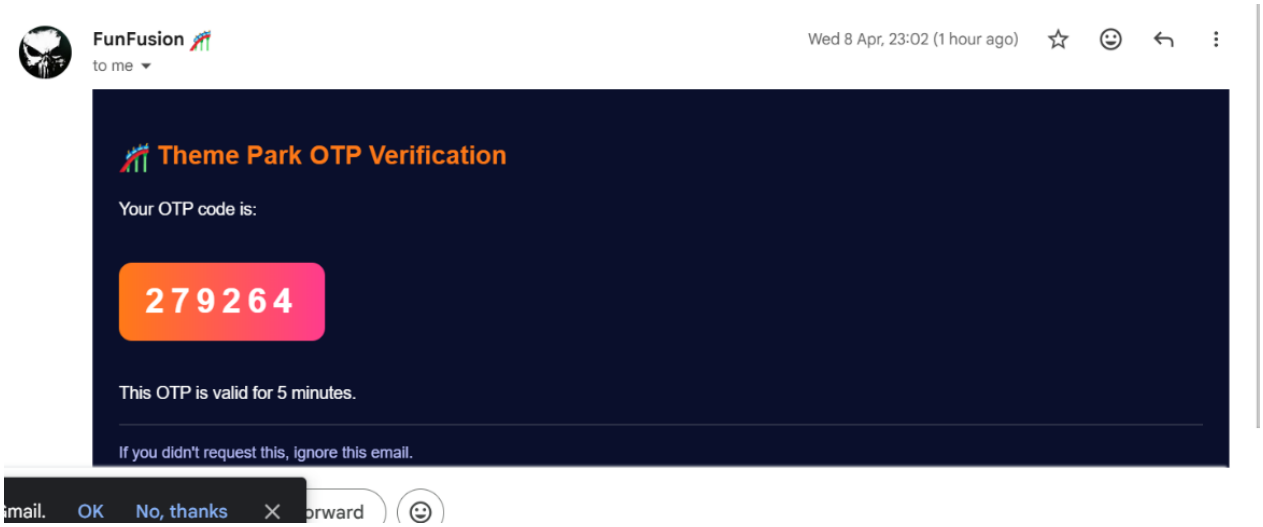
3.1 User enters email during signup
3.2 System generates 6-digit OTP
3.3 OTP sent via Nodemailer
3.4 OTP stored in temporary Map
3.5 User enters OTP
3.6 System verifies OTP
3.7 Account is activated

Test Cases:

TC ID	Scenario	Input	Expected Output
TC09	OTP sent	Valid email	OTP received
TC10	Correct OTP	Matching OTP	Verified successfully
TC11	Wrong OTP	Incorrect OTP	Error message

ScreenShots :





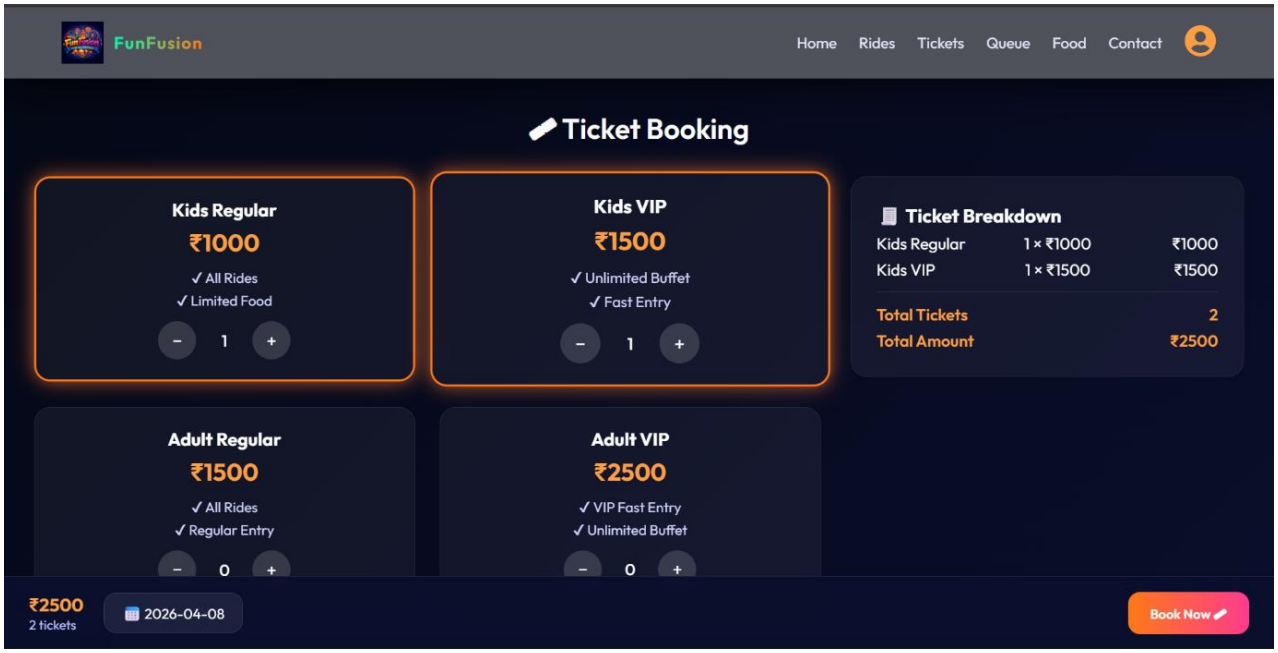
4. Ticket Booking Module

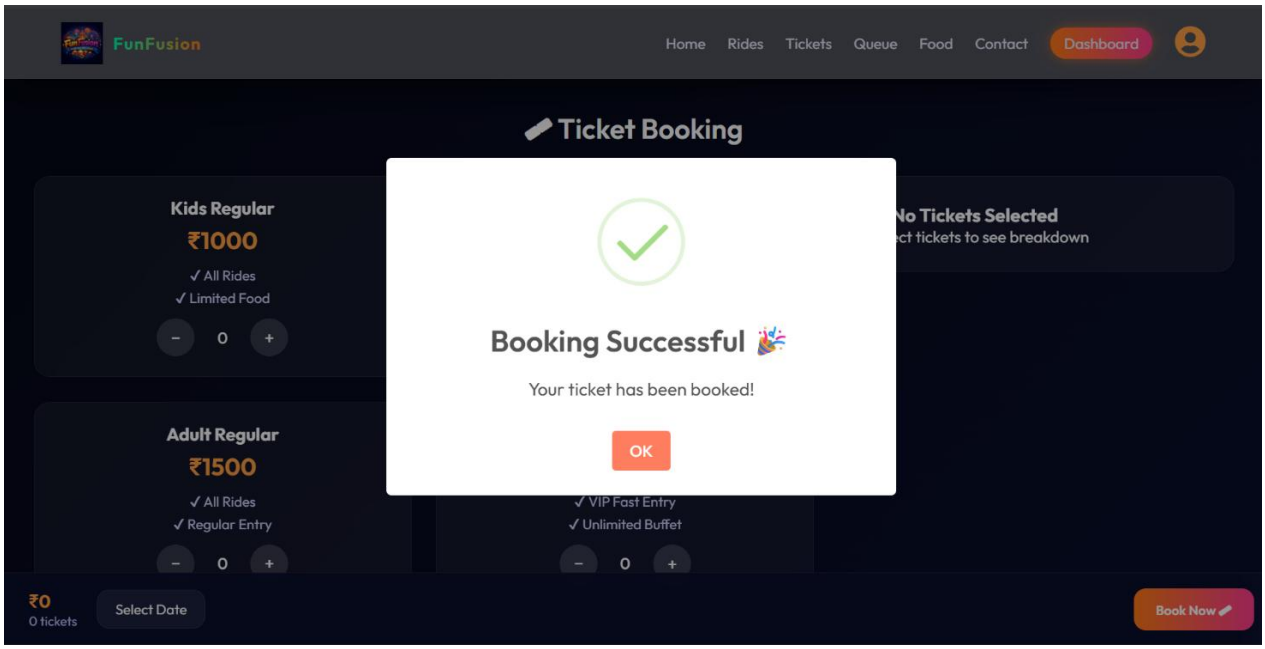
4.1 User selects ticket plan
4.2 User selects visit date
4.3 User selects quantity
4.4 System calculates total price
4.5 Booking request sent to backend
4.6 QR code generated
4.7 Booking stored in MongoDB

Test Cases:

TC ID	Scenario	Input	Expected Output
TC12	Book ticket	Valid details	Booking successful
TC13	Missing date	No date	Error message
TC14	QR generation	Booking success	QR displayed

ScreenShots :





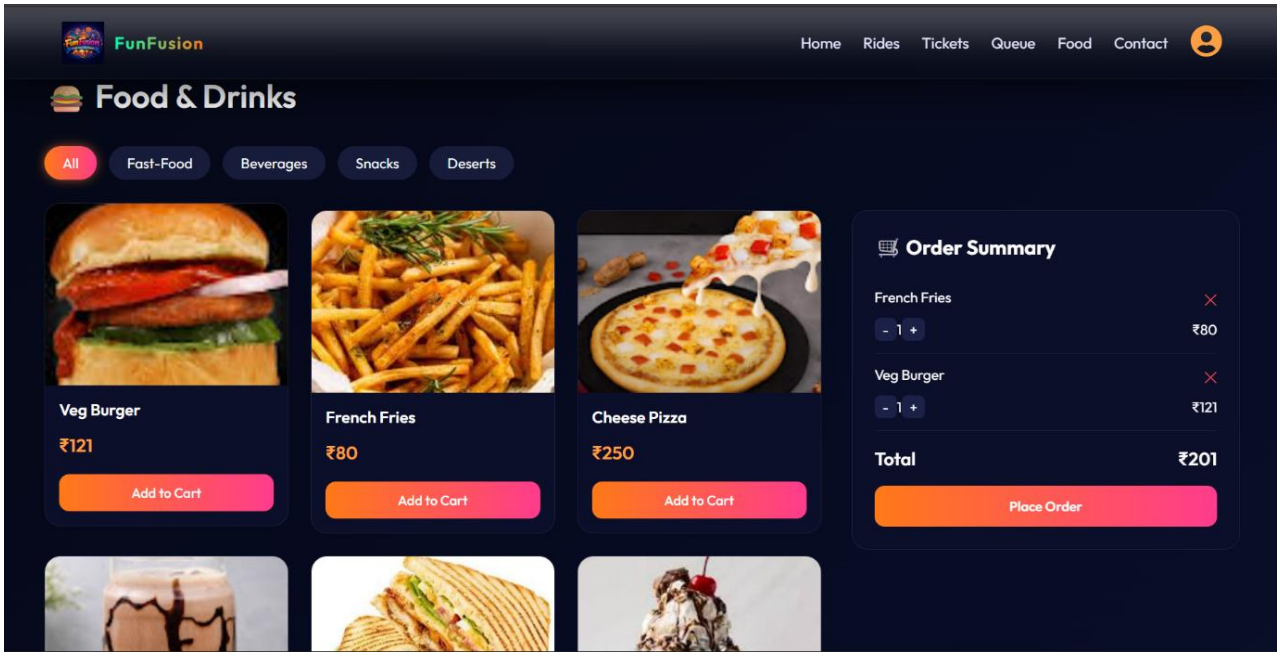
5. Food Ordering Module

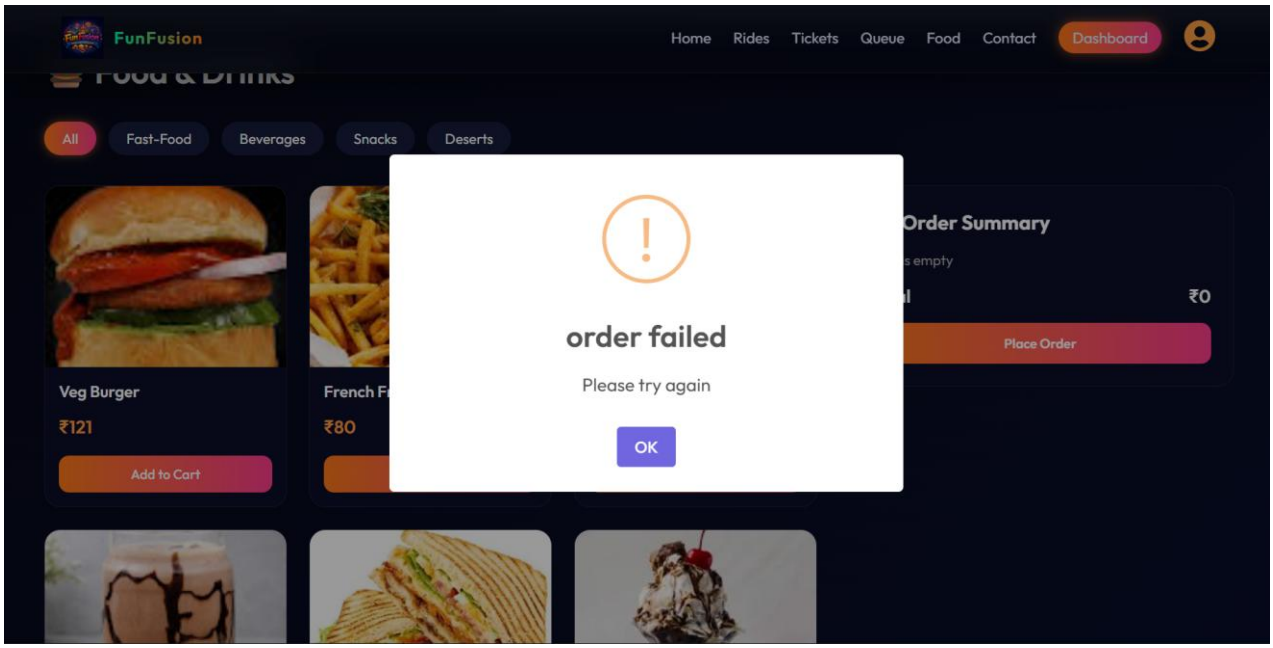
5.1 User views food items (categories: Fast-Food, Beverages, Snacks, Desserts)
5.2 User adds items to cart
5.3 User updates quantity
5.4 System calculates total amount
5.5 Order stored in MongoDB
5.6 QR receipt generated

Test Cases:

TC ID	Scenario	Input	Expected Output
TC15	Add food	Select item	Added to cart
TC16	Increase quantity	+ button	Quantity updated
TC17	Place order	Valid cart	Order success
TC18	Empty cart	No items	Error shown

ScreenShots :





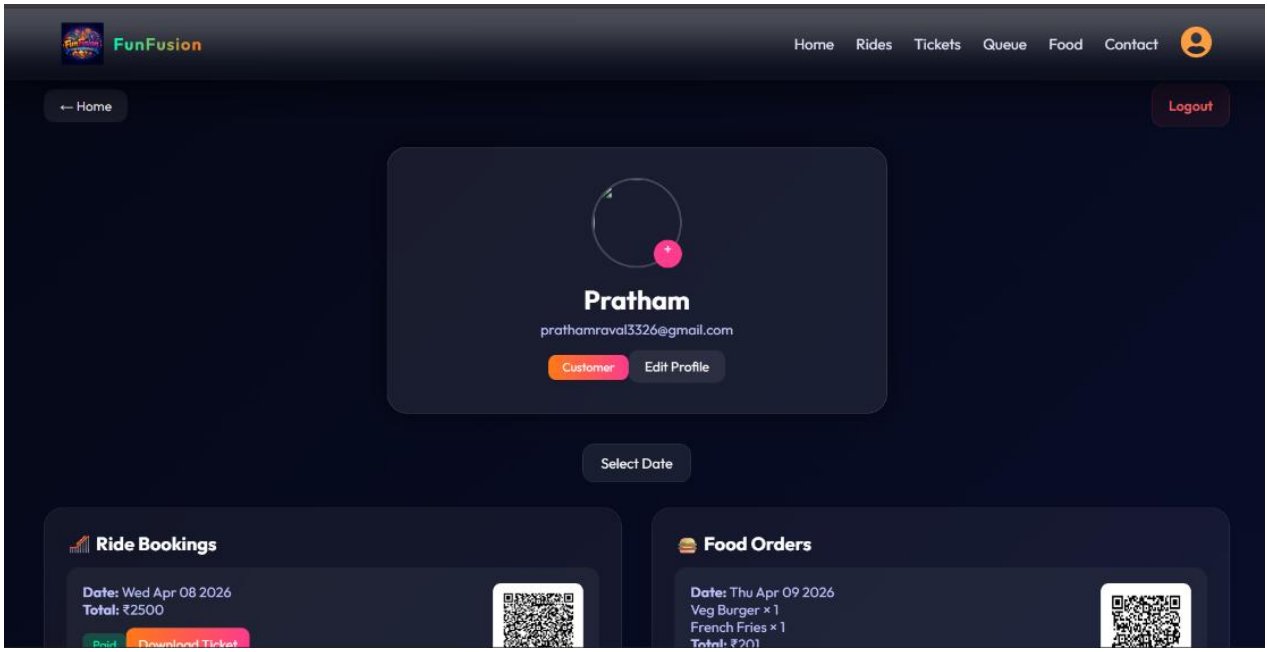
6. User Profile Module

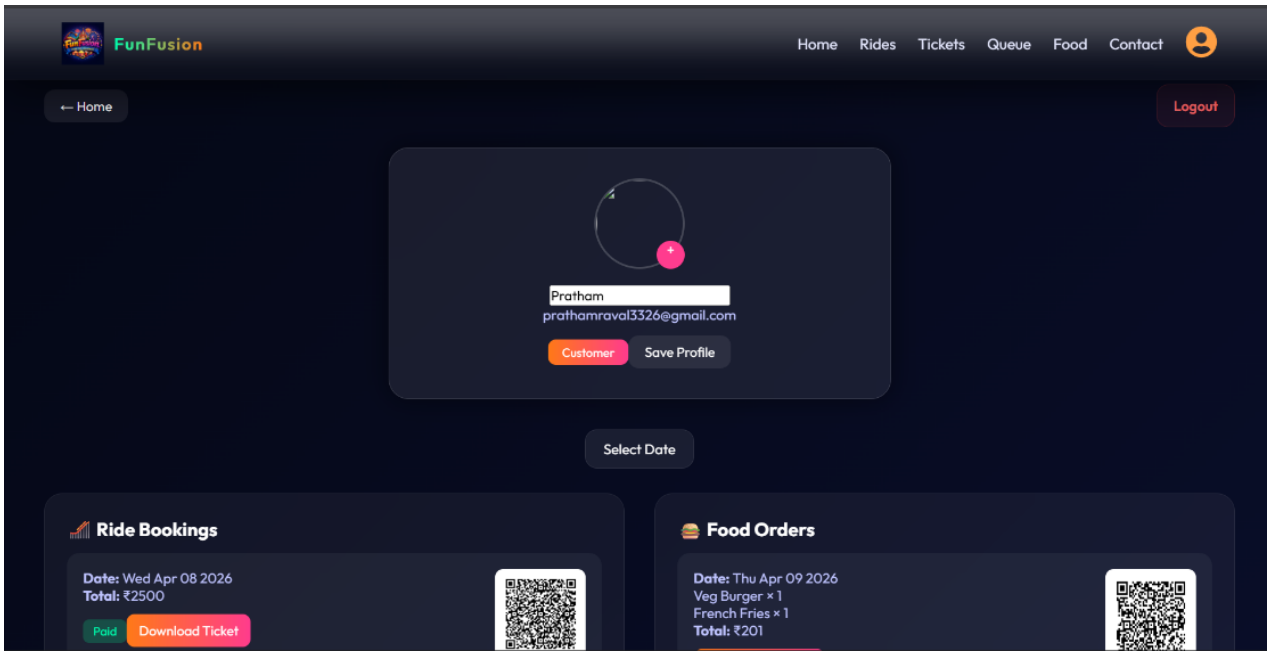
6.1 User logs in
6.2 System fetches booking history (by email)
6.3 System fetches food orders
6.4 Data displayed in profile
6.5 User can logout

Test Cases:

TC ID	Scenario	Input	Expected Output
TC19	View profile	Logged in user	Data displayed
TC20	Booking history	Existing bookings	List shown
TC21	Food orders	Existing orders	Orders displayed

ScreenShots :





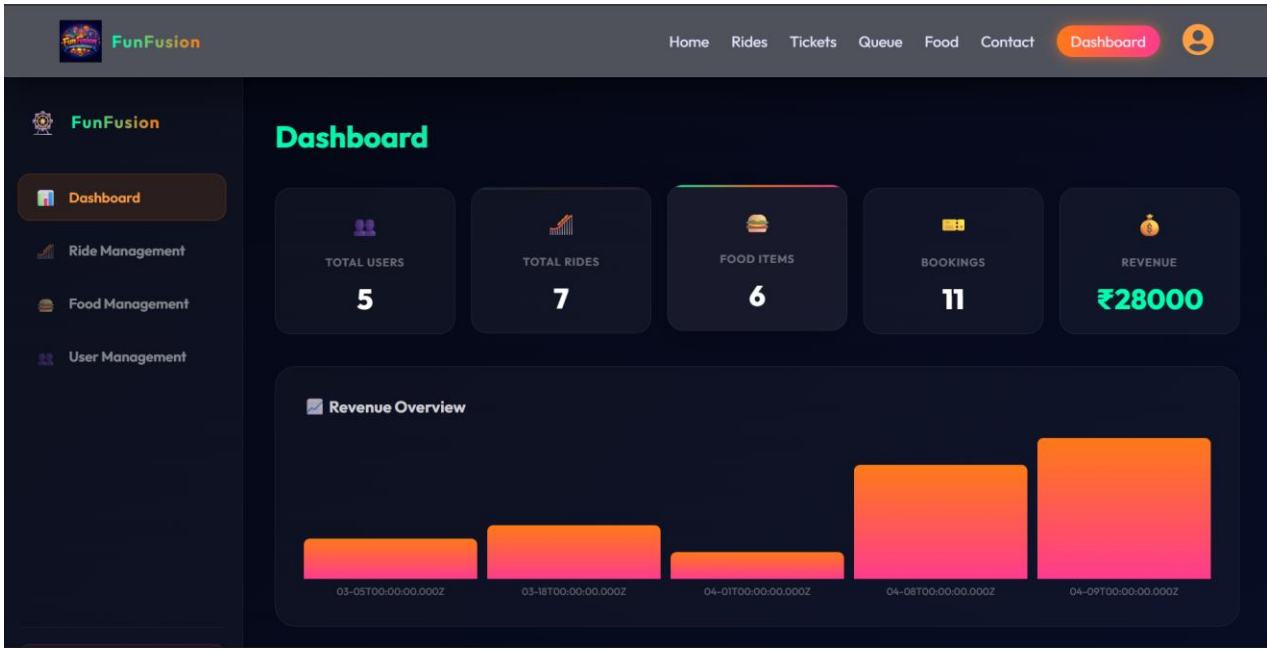
7. Admin Dashboard Module

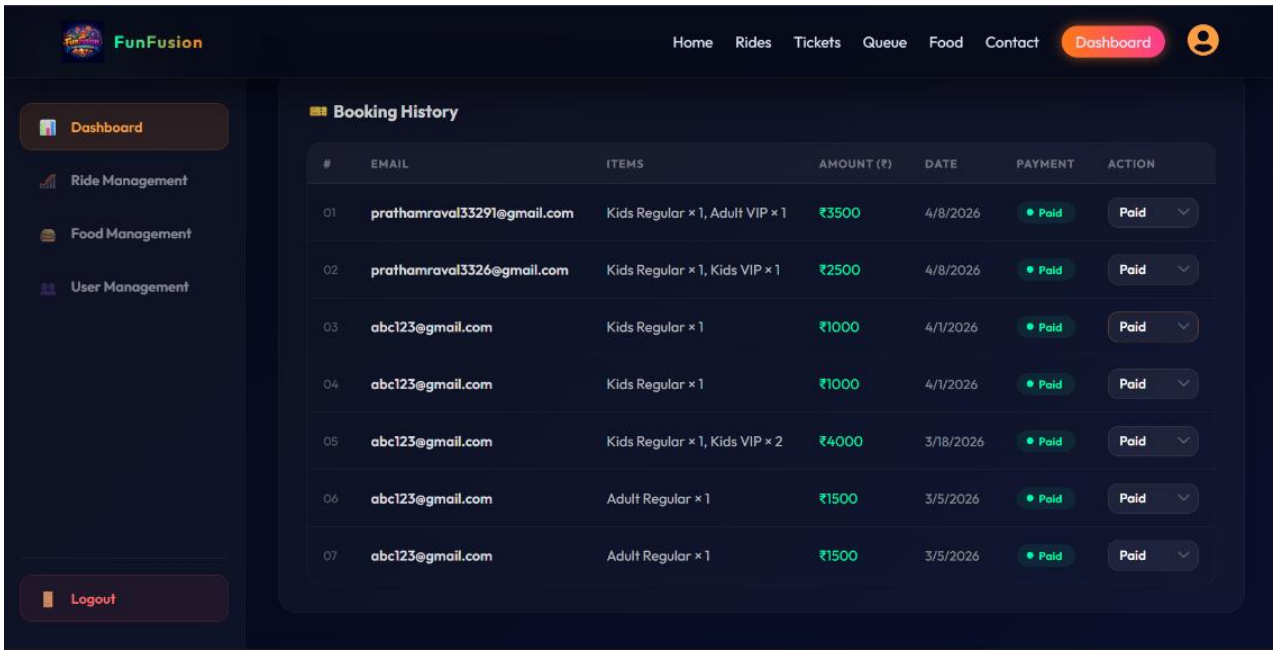
7.1 Admin logs in
7.2 Dashboard fetches system data
7.3 Stats calculated (users, bookings, revenue)
7.4 Admin sees Booking history
7.5 Admin updates Booking history

Test Cases:

TC ID	Scenario	Input	Expected Output
TC22	View dashboard	Admin login	Stats displayed
TC23	Data accuracy	DB data	Correct analytics
TC24	Update Booking history	Admin login	Stats displayed

ScreenShots :





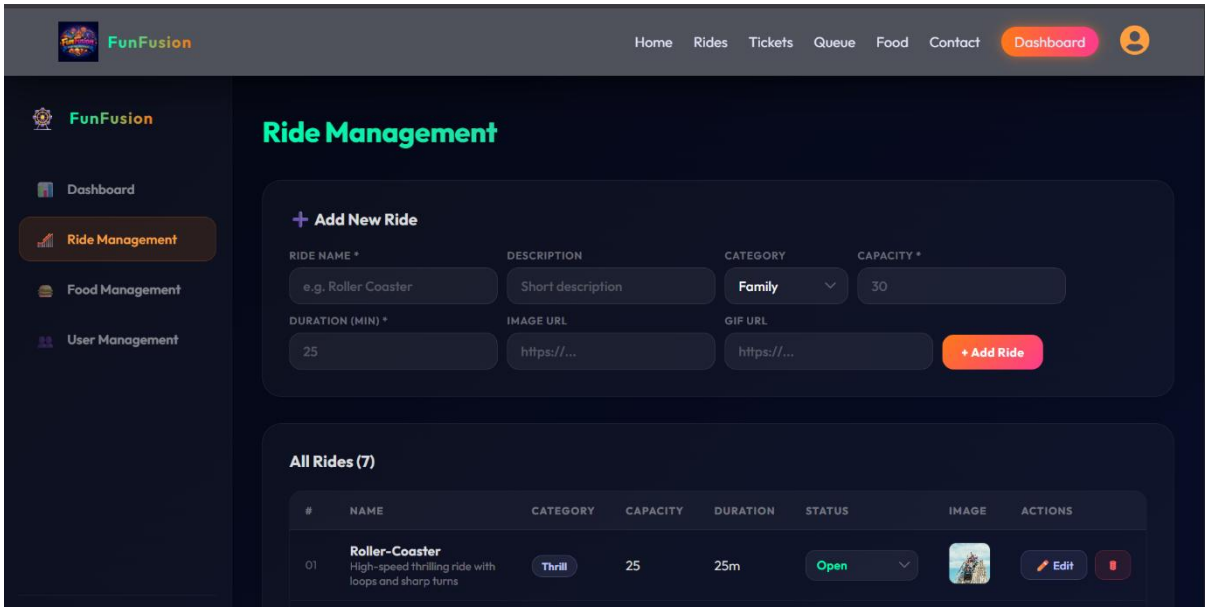
8. Admin Ride Management Module

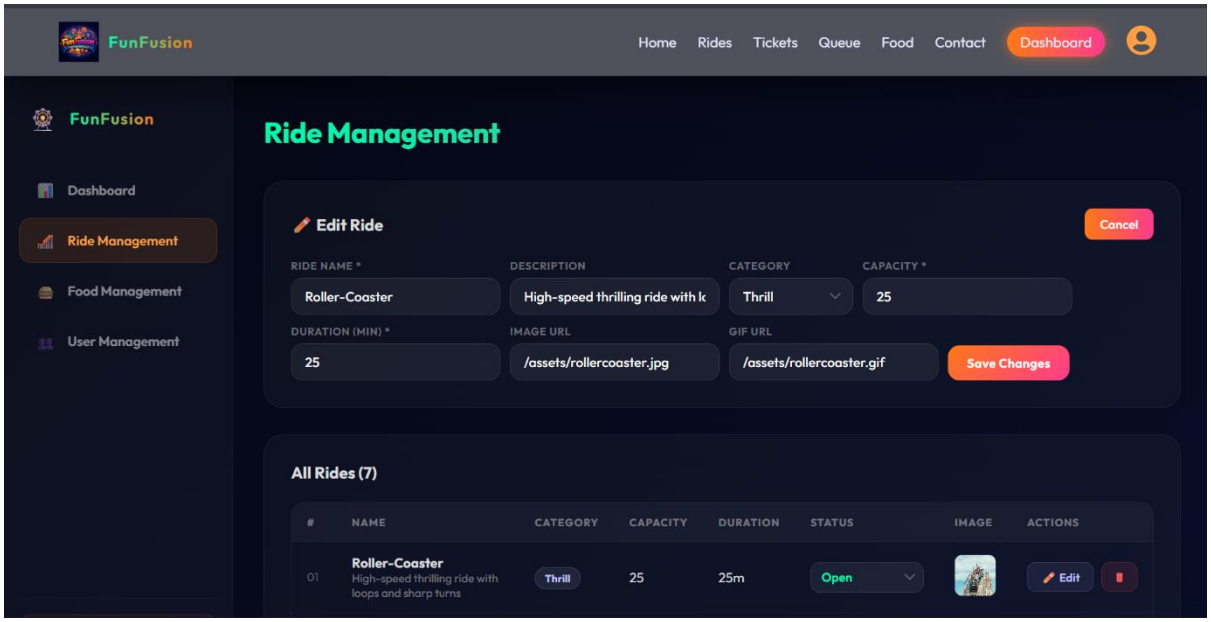
8.1 Admin logs in
8.2 Admin opens ride management page
8.3 Admin adds new ride
8.4 Ride data stored in MongoDB
8.5 Admin updates ride details
8.6 Admin deletes ride
8.7 Ride status updated dynamically

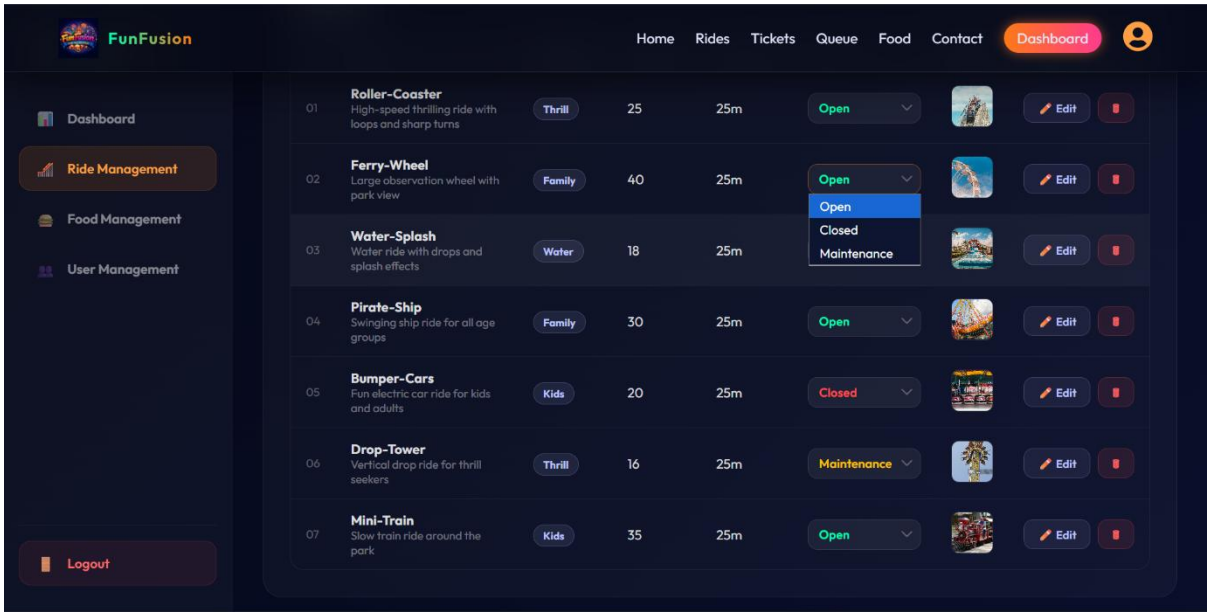
Test Cases:

TC ID	Scenario	Input	Expected Output
TC25	Add ride	Valid ride data	Ride added
TC26	Update ride	Modified data	Ride updated
TC27	Delete ride	Ride ID	Ride removed

ScreenShots :






FunFusion
Home
Rides
Tickets
Queue
Food
Contact
Dashboard
Dashboard
Ride Management
Food Management
User Management
Logout
01
Roller-Coaster
High-speed thrilling ride with loops and sharp turns
Thrill
25
25m
Open
Edit
02
Ferry-Wheel
Large observation wheel with park view
Family
40
25m
Open
Open
Closed
Maintenance
Edit
03
Water-Splash
Water ride with drops and splash effects
Water
18
25m
Edit
04
Pirate-Ship
Swinging ship ride for all age groups
Family
30
25m
Open
Edit
05
Bumper-Cars
Fun electric car ride for kids and adults
Kids
20
25m
Closed
Edit
06
Drop-Tower
Vertical drop ride for thrill seekers
Thrill
16
25m
Maintenance
Edit
07
Mini-Train
Slow train ride around the park
Kids
35
25m
Open
Edit

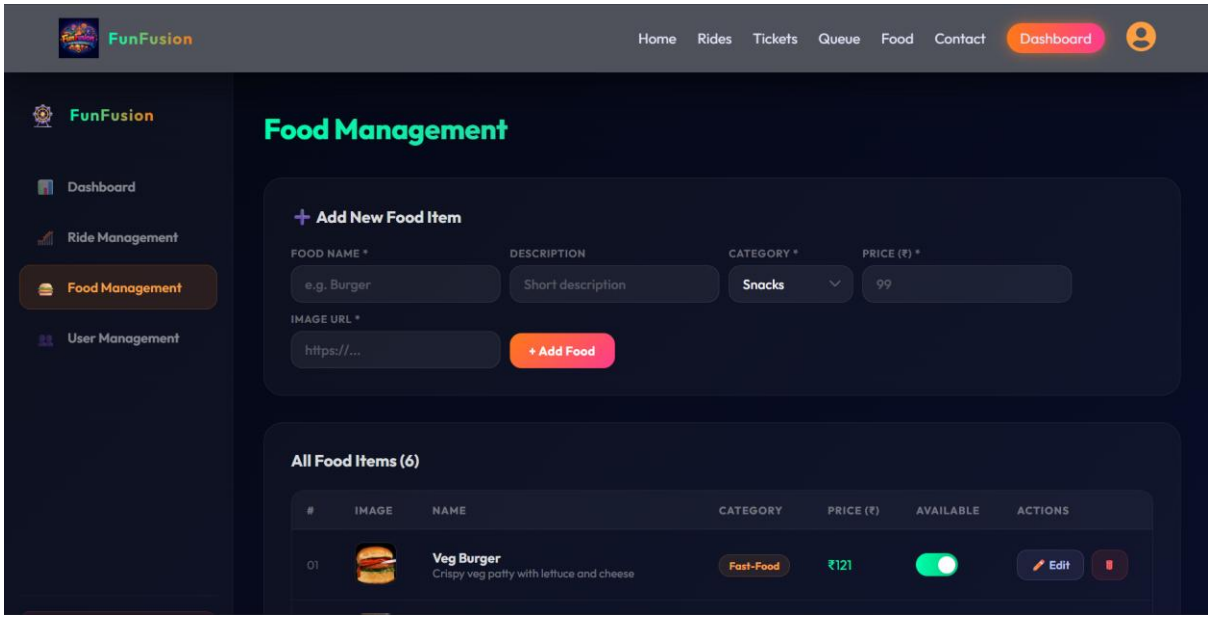
9. Admin Food Management Module

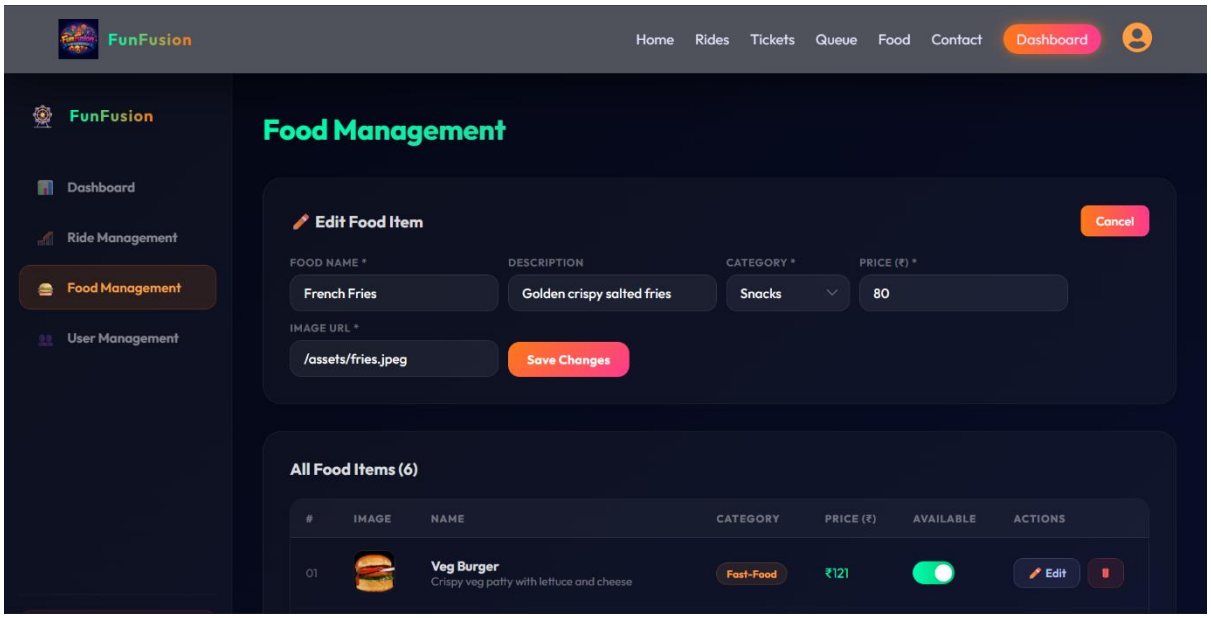
9.1 Admin opens food management page
9.2 Admin adds food item
9.3 Food categorized (Fast-Food, Beverages, etc.)
9.4 Admin updates price/details
9.5 Admin deletes item

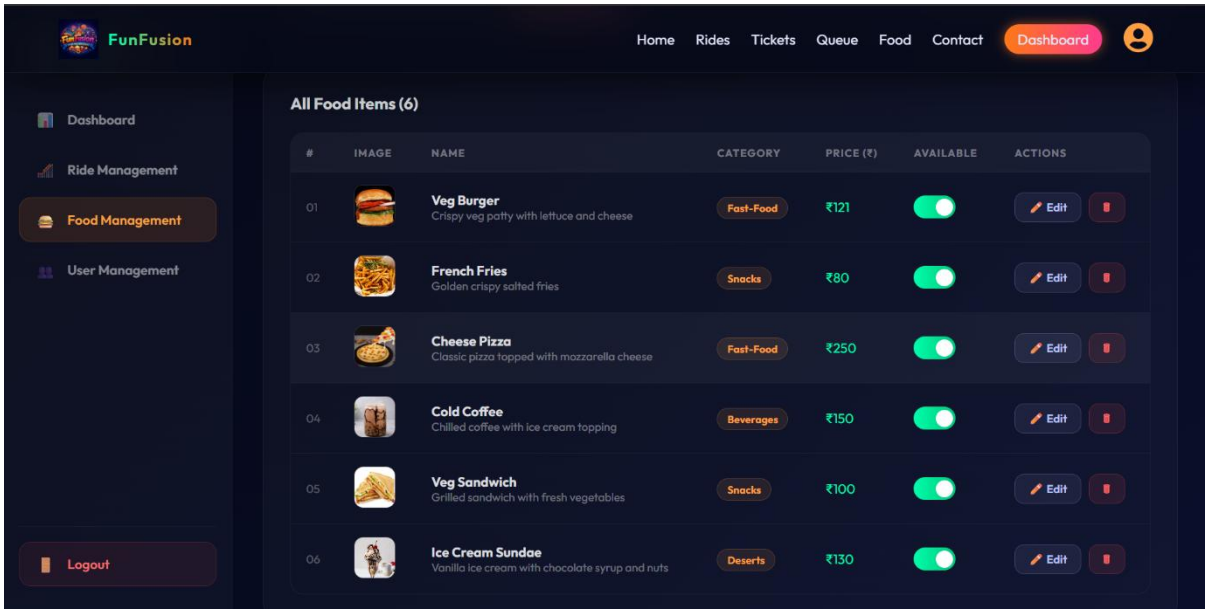
Test Cases:

TC ID	Scenario	Input	Expected Output
TC28	Add food	Valid data	Food added
TC29	Update food	New price	Updated successfully
TC30	Delete food	Food ID	Removed

ScreenShots :






FunFusion
Home
Rides
Tickets
Queue
Food
Contact
Dashboard
Dashboard
Ride Management
Food Management
User Management
Logout
All Food Items (6)
IMAGE NAME CATEGORY PRICE (₹) AVAILABLE ACTIONS
01 Veg Burger Crispy veg patty with lettuce and cheese Fast-Food ₹121 Edit
02 French Fries Golden crispy salted fries Snacks ₹80 Edit
03 Cheese Pizza Classic pizza topped with mozzarella cheese Fast-Food ₹250 Edit
04 Cold Coffee Chilled coffee with ice cream topping Beverages ₹150 Edit
05 Veg Sandwich Grilled sandwich with fresh vegetables Snacks ₹100 Edit
06 Ice Cream Sundae Vanilla ice cream with chocolate syrup and nuts Deserts ₹130 Edit

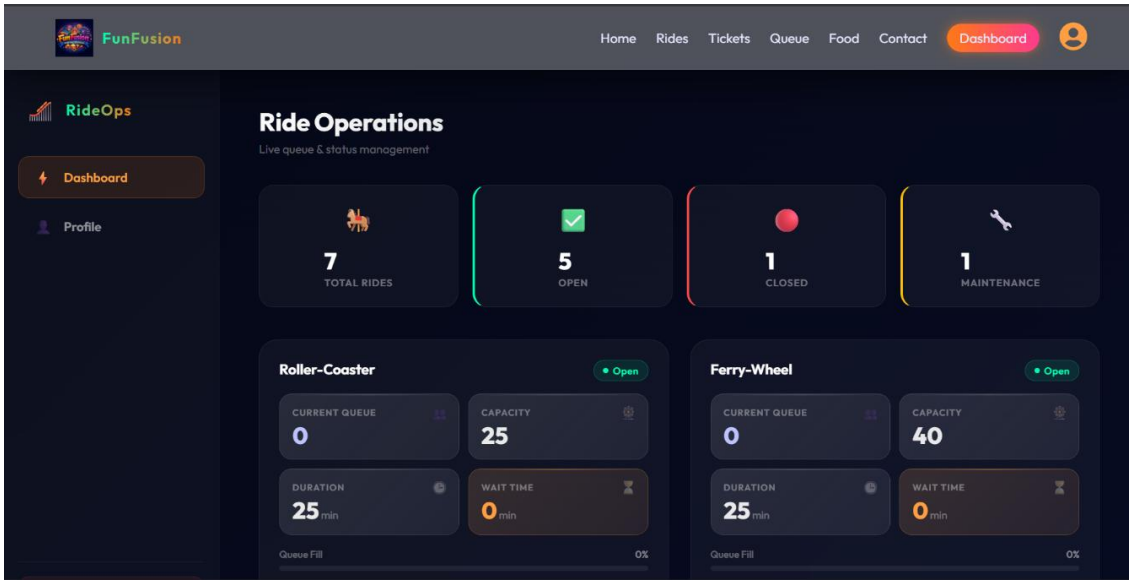
10. Ride Staff Module

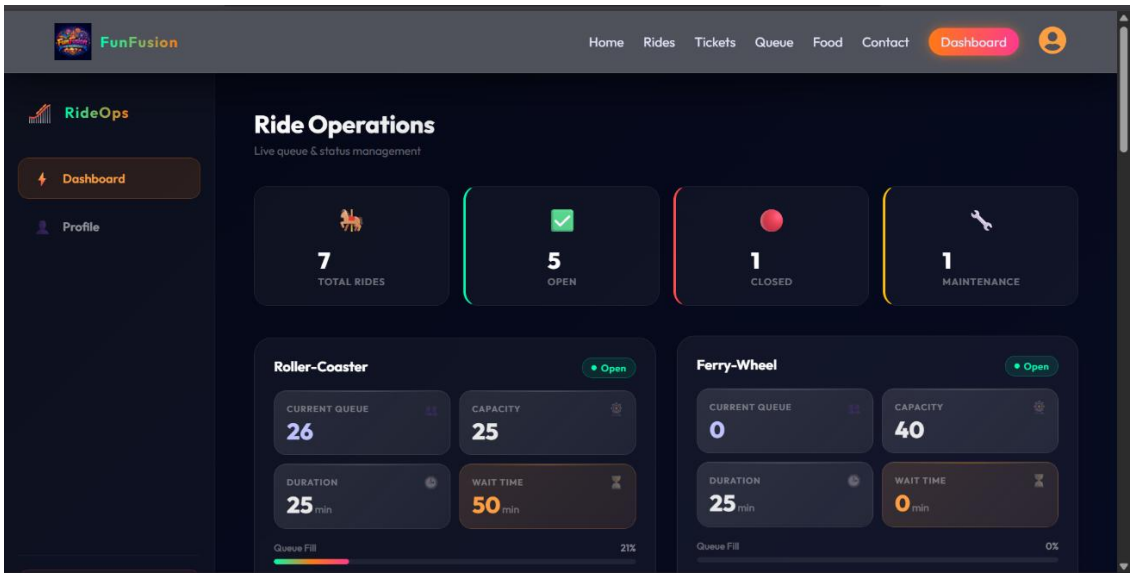
10.1 Ride Staff logs into the system
10.2 System verifies role as **RideStaff**
10.3 Ride Staff is redirected to Ride Staff Dashboard
10.4 Ride Staff can view assigned rides, current queue, estimated waiting time
10.5 Ride Staff updates ride status: Open Closed Maintenance
10.6 Ride Staff manages queue: Add users to queue, Remove users after ride cycle, Update queue count
10.7 System recalculates waiting time dynamically: Waiting Time = Queue × Average Duration
10.8 Real-time updates sent using Socket.io
10.9 All changes are stored in MongoDB

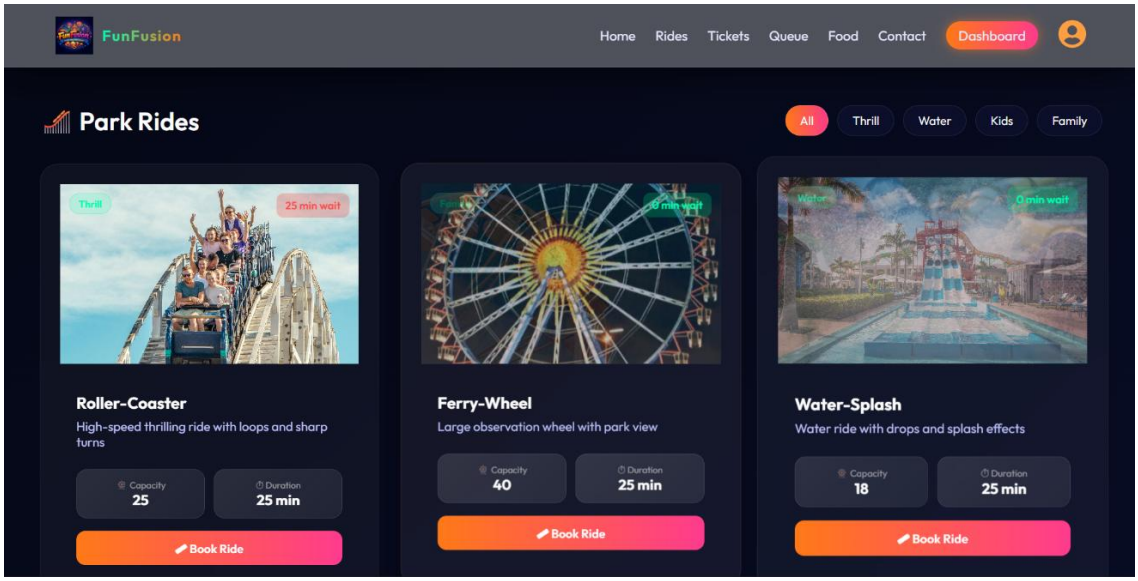
Test Cases:

TC ID	Scenario	Input	Expected Output
TC31	Ride staff login	Valid credentials	Dashboard opens
TC32	Role validation	RideStaff role	Access granted
TC33	View rides	Logged in staff	Assigned rides displayed
TC34	Update status	Open/Closed/Maintenance	Status updated
TC35	Add to queue	User count	Queue increases
TC36	Remove from queue	Ride cycle complete	Queue decreases
TC37	Waiting time calculation	Queue & duration	Correct time shown

ScreenShots :



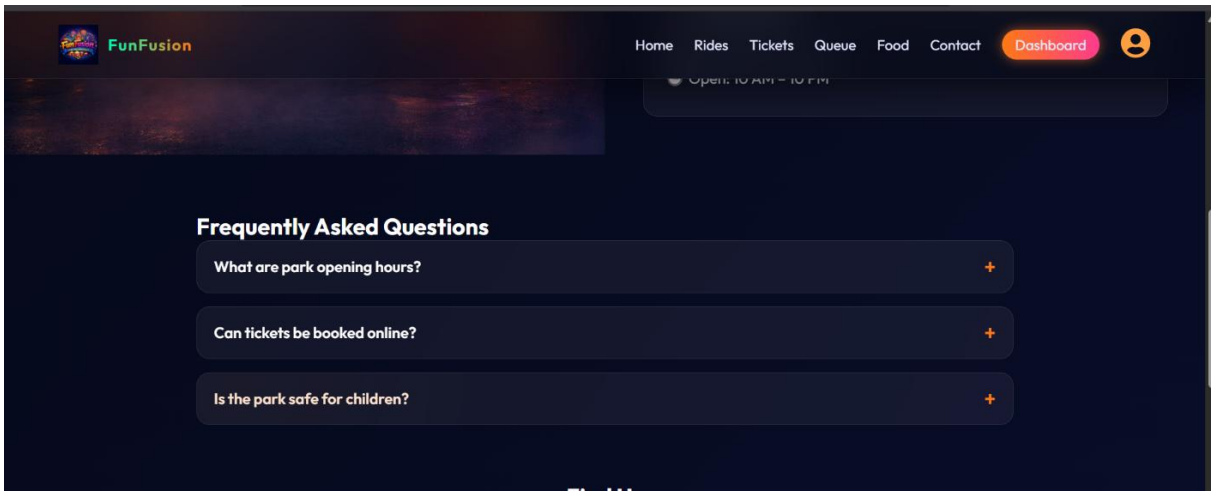
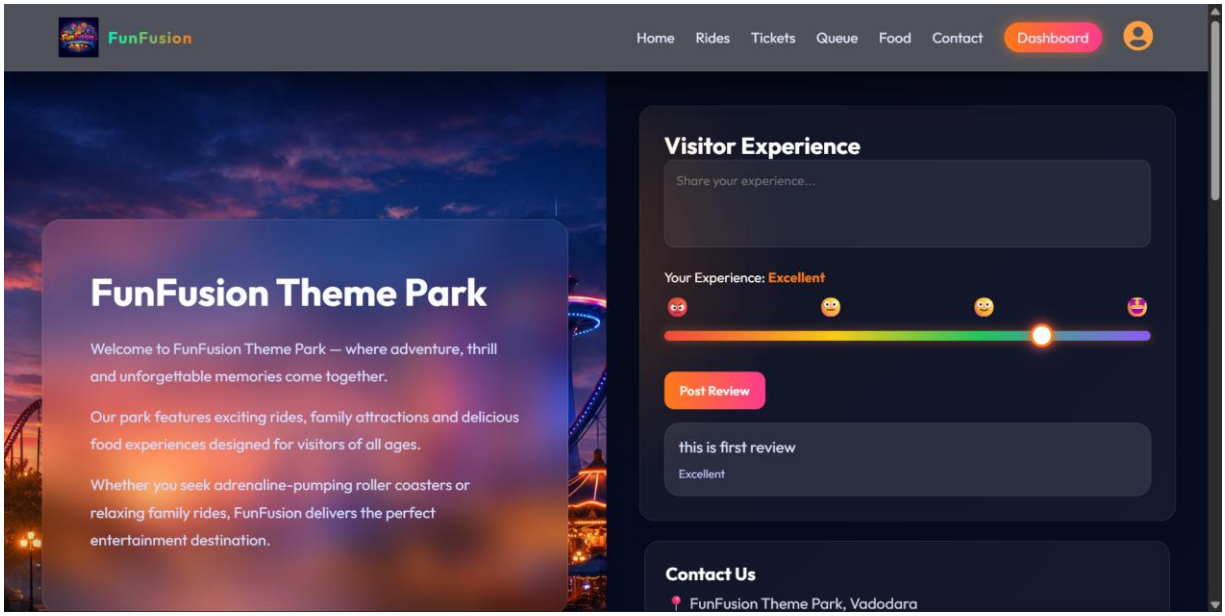
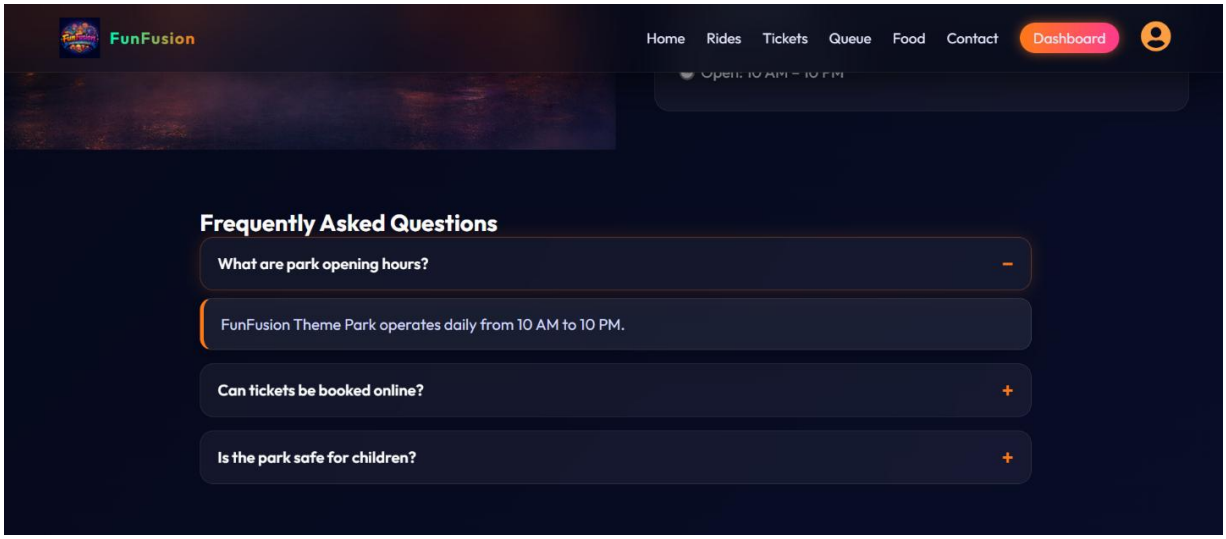




11.Contact Us Module

11.1 User opens Contact page
11.2 Left section shows theme park information with background image
11.5 User selects experience rating:Excellent (Purple), Good (Green), Average (Yellow), Not Enjoyed (Red)
11.6 User adjusts experience range meter (slider)
11.7 User submits form
11.8 System:Validates inputs Stores data in MongoDB Show's success message
11.9 FAQ section available:User clicks question Answer expands below
11.10 Location displayed:

Test Cases:

TC ID	Scenario	Input	Expected Output
TC38	Open contact page	User visits page	Page loads successfully
TC39	Submit contact form	Valid name, email, message	Message saved successfully
TC40	Missing fields	Empty inputs	Validation error shown
TC41	Select rating	Click rating option	Rating selected visually
TC42	Adjust experience slider	Move range bar	Value updates dynamically
TC43	Submit review	Valid data	Review stored in DB
TC44	FAQ expand	Click question	Answer expands
TC45	Location display	Open page	Location visible
TC46	Invalid email	Wrong format	Error message shown

ScreenShots :